

FactoryFlow (AECE Checkpoint) - Gap Analysis Review

AEC Electronics (Pty) Ltd | 2026-04-01

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Date: 2026-04-01 Reviewed by: Claude Code (automated analysis)

Project Overview

Stack: TypeScript monorepo - React/Vite frontend, Express.js backend, PostgreSQL + Drizzle ORM, Postmark email Deployment: Replit (autoscale), no Docker Company: AEC Electronics (Pty) Ltd, ~20 employees

There are 2 specs. Spec 1 (Leave & HR) is detailed and substantially implemented. Spec 2 (Payroll) is an empty stub - only the problem statement exists.

Spec 1: Leave Management & HR System

What's Working Well

Requirement	Status	Notes
REQ-001 Employee Data	Done	Rich schema: tax, POPIA, contract history, org hierarchy, manager
REQ-002 Leave Submission	Done	Date range, reason, document upload (JPEG/PNG/PDF as base64)
REQ-003 Annual Leave	Done	21 days/year, pro-rated accrual, 6-month carry-over expiry
REQ-005 Family Responsibility	Done	3 days/cycle, 4-month eligibility gate
REQ-009 Approval Workflow	Done (Enhance	Spec says Manager->HR; system implements Manager->HR->MD
REQ-010 Email Notifications	Mostly done	Submission, approval, rejection, stage transitions all notified

REQ-012 Balance	Done	Per-type balances with progress bars on employee dashboard
Display		
Critical Gaps (Must-Fix)		
1. REQ-006 - Statutory Special Leave: Not Implemented Maternity (4 months), Parental (10 days), Adoption, and Commissioning leave types don't exist. These are BCEA-mandated.		
2. REQ-013 - Validation Rules: Not Enforced Server-Side No server-side checks preventing: - Leave submission exceeding available balance		
3. REQ-004 - Sick Leave Probationary Rule: Incorrect The first-6-month rule (1 day per 26 days worked) is not applied - simple monthly pro-rating is used instead. The leave_rule_phases schema exists for this but the BCEA engine ignores it.		
4. REQ-008/011 - Unpaid Leave Notice & Discretion: Not Implemented No 7-day notice period validation, no short-notice warnings, no discretion logging for manager decisions.		
5. REQ-010 - Escalation Reminder Automation: Not Triggered The /api/leave-requests/send-escalation-reminders endpoint exists but there's no cron or scheduler. It must be manually called - the 12-hour auto-escalation from the spec never fires.		
6. REQ-014 - Audit Trail: Incomplete Only leave decision fields (approver, notes, timestamp) are tracked. No system-wide audit log table for profile edits, balance adjustments, or settings changes.		

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Medium Gaps (Should-Have)

7. REQ-007 - Custom Leave Rule Engine: Half-Done Leave rules and phases are configurable in the DB and admin UI, but the auto-calculation engine only handles 3 hardcoded BCEA types (Annual, Sick, FRL). Custom rules stored in leaveRules are never automatically applied to generate balances.
8. REQ-004 - Medical Certificate Rules: Not Enforced No flagging when sick leave exceeds 2 consecutive days, and no Fri/Mon/public-holiday proximity detection (per company policy).
9. REQ-015 - HR Reporting: Limited The HR dashboard shows basic counts. Missing: sick leave pattern/excessive absence alerts, leave trend analytics by type over time.
10. REQ-016 - Policy Configuration: Partially Hardcoded The 6-month carry-over grace and the Manager->HR->MD hierarchy are hardcoded - not configurable through the settings UI.

Spec 2: Payroll System

Status: 0% implemented. Spec itself is 0% complete.

The spec file contains only a problem statement and 4 stakeholder role names. No requirements, no user stories, no scope. Nothing in the codebase touches payroll - no PAYE/UIF/SDL, no tax tables, no IRP5/EMP201 generation, no payslips.

The taxNumber field on users and the companies table are minor incidental foundations but nothing more.

What Exists Beyond the Specs

The codebase has substantial features that are explicitly out of scope per spec 1 but fully built:

- Attendance/Time Tracking - Full clock-in/out, AWOL, infringement detection, auto-clock-out, bulk entry, PDF reports

- Facial
+ SSD

Infrastructure State

Aspect	Current State
Hosting	Replit autoscale
Database	PostgreSQL 16 (Replit module), 18 tables
Email	Postmark
Auth	localStorage-based (no server-side session validation)

CI/CD	None
Docker	None
Scheduler/Cron	None - escalation reminders can't auto-fire

Notable security concern: API endpoints do minimal authorization checking - role-based access is not consistently enforced server-side. Auth state lives entirely in localStorage.

Priority Recommendation

Priority	Item
P0	Server-side leave validation (balance, overlap, date range)
P0	Statutory special leave types (Maternity, Parental, etc.)
P1	Sick leave probationary accrual rule fix
P1	Escalation reminder cron (Replit scheduled jobs or external trigger)
P1	Audit log table
P2	Unpaid leave 7-day notice & discretion logging
P2	Custom leave rule execution engine
P2	Medical certificate flag triggers
P3	HR reporting / sick leave pattern alerts
P3	Server-side session auth
Backlog	Payroll (spec incomplete - needs requirements first)
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Full Requirement Traceability

Database Tables



users	Employee profiles	REQ-001
departments	Department management	Beyond spec
user_groups	Admin user grouping	Beyond spec
employee_types	Worker/contractor types	REQ-001 (partial)
leave_rules	Leave accrual configuration	REQ-007, REQ-016
leave_rule_phases	Tiered accrual phases	REQ-004, REQ-007
companies	Payroll company entities	Payroll spec (shell)
leave_balances	Leave balance tracking	REQ-003, REQ-012
leave_requests	Leave request workflow	REQ-002, REQ-009
attendance_records	Clock in/out records	Out of scope per spec
settings	System configuration KV	REQ-016
contract_history	Employment changes	REQ-001
grievances	Workplace grievances	Beyond spec
public_holidays	Holiday calendar	Beyond spec
notifications	In-app notifications	REQ-010
org_positions	Org chart hierarchy	Beyond spec
password_reset_tokens	Password reset flow	Beyond spec
face_descriptors	Multi-angle face data	Beyond spec

Client Pages

Page	Purpose
ModeSelect.tsx	Entry point: attendance kiosk vs employee portal
Login.tsx	Worker login (ID or face)
AdminLogin.tsx	Admin login (email/password or face)
Dashboard.tsx	Employee dashboard: balances, recent requests, clock status
LeaveRequest.tsx	Submit leave request with file upload
Attendance.tsx	Employee attendance history

AttendanceKiosk.tsx	Shared terminal face-recognition kiosk
AttendanceTileMode.tsx	Tile-based attendance view
AdminDashboard.tsx	Admin shell with sidebar navigation to 14 sections
MaintainerDashboard.tsx	Limited admin for data entry
ResetPassword.tsx	Password reset form
EmployeeProfile.tsx	Employee self-view profile
OrgChart.tsx	Visual org chart
Grievances.tsx	Employee grievance submission
LeaveCalendar.tsx	Calendar view of approved leave
AttendanceReports.tsx	Attendance reporting/analytics
not-found.tsx	404 page

